

Seongwon Jang

jungseonglian@gmail.com | [jungseong.github.io](https://github.com/jungseong) | [linkedin.com/in/seongwonjang](https://www.linkedin.com/in/seongwonjang) | github.com/JungSeong

TECHNICAL SKILLS

Languages: C, C++, Python

Frameworks & Libraries: PyTorch, TensorFlow, vLLM, DeepSpeed, LangChain, OpenCV, LeRobot, Hugging Face, Weights & Biases, MCP, FastAPI

DevOps & Tools: Docker, CMake, Pixi

Middleware: ROS2

Platforms: Linux

SCM: Git

Databases: PostgreSQL, MySQL

EXPERIENCE

Research Intern

Jul. 2025 – Sep. 2025

Ubiquitous AI Lab (UAI Lab), UNIST (Advisor: Prof. Taesik Gong)

Ulsan, Korea

- Investigated parameter-efficient fine-tuning (PEFT) methods for Mixture-of-Experts (MoE) model
- Benchmarked LoRA, ESFT, and the proposed technique on **OLMoE-1B-7B-Instruct** across major NLP benchmarks to evaluate fine-tuning efficiency and task performance.

CONTESTS

🏆 AI for Industry Challenge | PyTorch, ROS2, YOLOv8, LeRobot

Apr. 2026 – May. 2026

- Developed a UR5e cable-insertion policy for SFP/SC ports by integrating ROS2 policy nodes with the AIC Gazebo engine
- Built a multi-view perception pipeline for port localization: automated data collection/labeling code, trained a port detector, and fused detections via stereo triangulation
- Implemented a staged motion policy combining YOLO-based approach, quintic waypoint interpolation, pose-prediction-based yaw/XY alignment, and force/torque guarded insertion retries.
- Organized project-wide ROS2 packages and model/data infrastructure for LeRobot ACT training, Hugging Face dataset/model upload, and randomized simulator evaluation.

🏆 Structural Stability Physical Reasoning AI | PyTorch, VideoMAE, Knowledge Distillation

Mar. 2026 – Mar. 2026

- Ranked 66th out of 484 teams** (top 17.7%).
- Identified and resolved flaws in the existing data augmentation pipeline
- Validated the effectiveness of video frame-based augmentation through ablation studies.
- Selected optimal hyperparameters (model architecture, learning rate, weight decay) via systematic sweep experiments.
- Extracted feature vectors from video data using a **VideoMAE** encoder and designed regularization functions to improve generalization.
- Selected a knowledge distillation model and designed corresponding regularization functions for model compression.

MAJOR PROJECTS

🏆 Advanced Q&A Chatbot System for Public Offices | LLM, vLLM, DeepSpeed, LoRA

Jan. 2026 – Mar. 2026

- Accelerated inference and training using **vLLM** and **DeepSpeed**.
- Containerized the system with Docker Compose and integrated MCP for improved accessibility.
- Constructed high-quality synthetic Q&A datasets using Chain-of-Thought (CoT) prompting and DeepEval Synthesizer to secure golden datasets for fine-tuning.
- Applied SFT with **LoRA** to improve instruction-following performance.
- Deployed trained adapters and tokenizers to the HuggingFace Hub.

🏆 RoboDine: AI-Based Robotic Restaurant Automation Platform | ROS2, YOLO, OpenCV

Apr. 2025 – Jun. 2025

- Enhanced robot arm grasping by refining 6-DoF pose estimation with **YOLO-v8-pose** (Dropout + Pose Loss tuning) and a 5-frame moving average to suppress noisy coordinate predictions; applied **solvePnP** for full 6-DoF pose recovery.
- Implemented a foreign object detection algorithm for the robot arm using **Hough Circle Transform** and **DBSCAN** clustering.
- Built an obstacle type classification and avoidance pipeline for the mobile robot using Mediapipe; developed a ROS2 package publishing obstacle type, LiDAR distance, and robot ID as topics.
- Practiced Agile development with Git, Jira, and Confluence in a team of 6.

MINOR PROJECTS

🏆 Robo – Household Companion | Raspberry Pi 4, Gemini API

Mar. 2025 – May. 2025

- Developed a human-robot interactive home companion on Raspberry Pi 4; integrated microphone, touch sensors, and **Gemini API** for user intent recognition and action execution.

🏆 AIBAK: AI-Based Children's Book Kiosk Service | GPT-4o-mini, LangChain, ElevenLabs

Sep. 2024 – Dec. 2024

- Implemented AI-driven story and title generation using **ChatGPT-4o-mini** and **LangChain**; added TTS narration via ElevenLabs API.
- Awarded **Excellence Prize** at the In-Jeju IoT Innovation Challenge.

🏆 Facial Expression Recognition Web App | Residual Masking Network, Flask, MySQL

Apr. 2024 – May. 2024

- Led a team to build a web-based facial expression recognition app using **Residual Masking Network**; developed the **Flask** + **MySQL** backend and managed the schedule via WBS.

EDUCATION

Sejong University <i>B.S. in Electronics and Information Engineering</i> GPA: 4.02 / 4.5 (Major GPA: 4.13 / 4.5), Magna Cum Laude	Seoul, South Korea <i>Mar. 2019 – Feb. 2026</i>
--	--

HONORS & AWARDS

2025 – Grand Prize, ROS2 & AI Autonomous Driving / Robot Arm Developer Bootcamp
2024 – Excellence Prize, 1st In-Jeju IoT Innovation Challenge
2024 – Encouragement Award, Sejong University Reading Contest (May 2024, Dec. 2023); Grand Prize (Oct. 2023)

CERTIFICATES

2025 – Engineer Information Processing (Dec. 2025)

ENGLISH

2026 – OPIc IH
2025 – TOEIC 925

TUTOR EXPERIENCE

 C Language Algorithm Mentor	Mar. 2024 – Dec. 2024
- Designed and authored algorithm problems in C, covering core data structures such as stacks, queues, and linked lists. - Provided code reviews and optimized solutions for students, focusing on time complexity and memory efficiency.	

ALGORITHM

 Problem Solving	Jun. 2025 – Present
Achieved Gold 1 rank on Baekjoon Online Judge (solved over 430+ problems).	